Downtime Log & Pareto

Step-by-step guide: log every stop and see which causes cost the most time

What it does

You record each stop with a reason. The **Pareto** sheet ranks the reasons by minutes lost, with each reason's share and the running total, so you can see the few causes behind most of your downtime. Filter by date range and machine.

Yellow cells: type here.	Grey cells: calculated. Don't type over them.

The sheets

Sheet	What it's for
How to use	A one-page summary.
Setup	Your machines (up to 30) and your downtime reasons with categories (up to 30).
Downtime Log	One row per stop. Up to 1,000 rows.
Pareto	Reasons ranked by minutes lost, with a Pareto chart of the top 12.
By Machine	Downtime minutes and stop counts per machine.

Before you start

1. Open the file in your spreadsheet app (see **Works with** in the listing). If Excel shows a yellow *Protected View* bar because the file was downloaded, click **Enable Editing**.
2. Save a copy under your own name (for example, add your plant or year) and keep the original download as a clean spare.
3. Read the **How to use** sheet. It's a one-page summary of this guide.

Step 1: Set up machines and reasons

1. Go to the **Setup** sheet.
2. Under **Machine**, list your machines, one per row.
3. Under **Downtime reason** and **Category**, edit the 12 starter reasons to match how your plant talks about stops. Keep the list short, about 10 to 15 reasons. Categories (Breakdown, Material, and so on) can be renamed too.

Setup

Machines and downtime reasons used by the drop-downs.

Machine
Press 01
CNC Lathe 02
Packaging Line A

Downtime reason	Category
Mechanical breakdown	Breakdown
Electrical / controls fault	Breakdown
Tooling failure	Breakdown
Changeover / setup	Setup & adjustment
Adjustment / tuning	Setup & adjustment
Waiting for material	Material
Waiting for upstream process	Material
Quality hold / inspection	Quality
No operator available	Labor
Planned maintenance	Planned
Meeting / training	Planned
Other (describe in notes)	Other

Setup: machines on the left, reasons and categories on the right.

Step 2: Log every stop

1. Go to the **Downtime Log** sheet and use one row per stop.
2. Enter **Date**, **Shift** and **Machine** (from the drop-down).
3. Enter **Start** and **End** as times, like **7:15** and **7:50**. A stop that crosses midnight, like 23:40 to 0:25, is handled automatically (45 minutes).
4. If you only know how long it lasted, leave Start and End blank and type the minutes in **Minutes (if no start/end)**. Typed minutes always win over start and end times.
5. Pick a **Reason** from the drop-down. **Duration**, **Category** and **Check** fill in by themselves. Check should say **OK**; otherwise it tells you what's missing, such as a date, a reason, or minutes that aren't a number of 0 or more.
6. Add **Notes** if useful, especially for "Other".
7. The rows marked **EXAMPLE** show the expected format. When you're ready, select just their **yellow** cells and press **Delete** to clear them. Don't delete whole rows: that removes the formulas in the grey columns too.

Downtime Log

One row per stop. Use start/end times, or type minutes if you only know the duration.

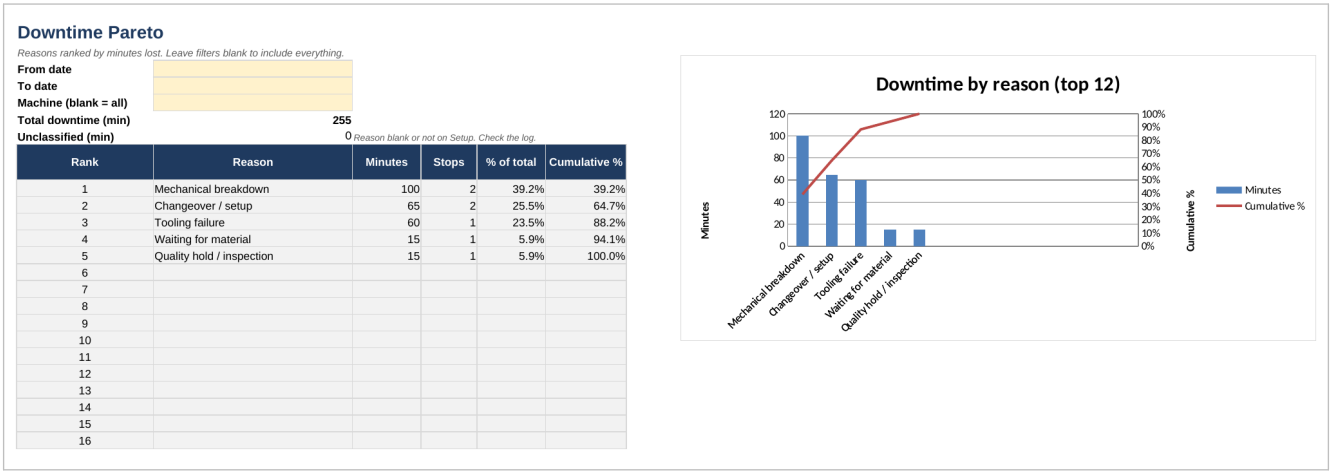
Date	Shift	Machine	Start (hh:mm)	End (hh:mm)	Minutes (if no start/end)	Duration (min)	Reason	Category	Notes	Check
2026-09-01	Day	Press 01	07:15	07:50		35	Changeover / setup	Setup & adjustment	EXAMPLE - Die change	OK
2026-09-01	Day	Press 01	10:05	10:20		15	Waiting for material	Material	EXAMPLE	OK
2026-09-01	Day	CNC Lathe 02	09:00	10:00		60	Tooling failure	Breakdown	EXAMPLE - Insert broke	OK
2026-09-01	Night	Packaging Line A	23:40	00:25		45	Mechanical breakdown	Breakdown	EXAMPLE - Conveyor jam, crossed midnight	OK
2026-09-02	Day	Press 01			55	55	Mechanical breakdown	Breakdown	EXAMPLE - Minutes typed directly	OK
2026-09-02	Day	Packaging Line A	13:00	13:30		30	Changeover / setup	Setup & adjustment	EXAMPLE	OK
2026-09-02	Day	CNC Lathe 02	14:10	14:25		15	Quality hold / inspection	Quality	EXAMPLE	OK

Downtime Log with the example rows, including a stop that crosses midnight.

If **Category** shows **Not in Setup**, the reason was typed instead of picked, or is spelled differently from the Setup list. Its minutes still count in the total, under **Unclassified** on the Pareto sheet. Pick the reason from the drop-down so it's counted under the right reason. A row counts as a **stop** when its duration is more than 0 minutes; rows without a date, or with invalid minutes, are left out of the totals.

Step 3: Read the Pareto

1. Go to the **Pareto** sheet.
2. Optional: set **From date**, **To date** and **Machine**. Leave any of them blank to include everything. Both dates are included in the range.
3. **Total downtime** is every valid logged minute in the range. **Unclassified** shows minutes whose reason is blank or not on the Setup list. Fix those in the log so they're ranked.
4. Read the table from the top: **Rank**, **Reason**, **Minutes**, **Stops**, **% of total** and **Cumulative %**. Reasons with no minutes in the range are left out.
5. The chart shows the top 12 reasons as bars, with the cumulative % as a line on the right-hand axis.
6. Start improvement work with the reasons at the top. Often the first two or three account for most of the lost time.



Downtime Pareto

Reasons ranked by minutes lost. Leave filters blank to include everything.

From date	
To date	
Machine (blank = all)	
Total downtime (min)	255
Unclassified (min)	0 *Reason blank or not on Setup. Check the log.*

Rank	Reason	Minutes	Stops	% of total	Cumulative %
1	Mechanical breakdown	100	2	39.2%	39.2%
2	Changeover / setup	65	2	25.5%	64.7%
3	Tooling failure	60	1	23.5%	88.2%
4	Waiting for material	15	1	5.9%	94.1%
5	Quality hold / inspection	15	1	5.9%	100.0%
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					

Pareto: reasons ranked by minutes, with cumulative % and the Pareto chart.

Step 4: Compare machines

The **By Machine** sheet shows total downtime, number of stops, average minutes per stop and each machine's share of all downtime, plus **No machine / not in Setup** and **All logged downtime** rows so the totals always add up. It uses the date range set on the Pareto sheet; the machine filter doesn't apply here.

Downtime by Machine

Uses the date range set on the Pareto sheet (machine filter is ignored here). A stop is any row with more than 0 min

Machine	Downtime (min)	Stops	Avg min per stop	% of all downtime
Press 01	105	3	35.0	41.2%
CNC Lathe 02	75	2	37.5	29.4%
Packaging Line A	75	2	37.5	29.4%

By Machine: downtime minutes, stops and share per machine.

Tips

- If **Other** climbs into your top reasons, read its notes and add the missing reason to Setup.
- Durations are rounded to whole minutes.
- Log stops the same day, while people remember the cause.

Questions

The log is full (1,000 rows). What now? Save the workbook as a new file for the next period, clear the yellow cells in the log, and keep going.

Can I rename the sheets or move columns? Please don't. The formulas refer to them by position.

Need help?

Message us through Etsy (the **Message seller** button on our shop or your order). Tell us which file, which sheet, and what you typed. We answer questions about using the templates as described in this guide; we can't do custom changes or set-up work.

These templates were designed with help from AI tools and checked by the seller.